Ashesh Kaji

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EDUCATION

MS in Computer Engineering
01/2026 – 12/2027 (expected)

NYU Tandon School of Engineering

Coursework: System Optimization Methods, Computer Systems Architecture, Special Topics in Computer Engineering. Research focus: portfolio optimization as layered control, convex optimization, reinforcement learning for financial decision-making.

BS with Honors in Cognitive Science
09/2021 – 06/2025

Specialization in Machine Learning and Neural Computation · UC San Diego

Honors thesis on statistical modeling of environmental exposures and neuroimaging data. Coursework: Deep Learning, Reinforcement Learning, Probability & Statistics, Linear Algebra, Algorithms.

WORK EXPERIENCE

Consulting AI Engineer
01/2026 – Present

SageX Global · Remote

- Architect and deploy production AI infrastructure: LLM serving pipelines, MCP (Model Context Protocol) servers, and agent orchestration systems with LangChain and AI-SDK
- Design custom data transformation and fine-tuning strategies across multi-modal data lifecycle requirements
- Build and maintain ACP-compliant agent communication protocols for distributed AI systems

LangChain · AI-SDK · ACP · MCP Servers · Python · Rust · Docker · Cloudflare Workers · Azure · AWS

Artificial Intelligence Engineer
09/2024 – 01/2026 · 1 year full-time

SageX Global · Remote

- Led development of multiple production AI products: LLMs, small language models (SLMs), statistical mapping models, and semantic database memory retrieval systems

- Designed end-to-end data pipelines for fine-tuning and deploying models with rigorous evaluation methodology
- Built RAG systems integrating vector databases with serverless deployment on Azure and AWS

PyTorch · Scikit-Learn · Azure · AWS · MLOps · NLP · RAG · LLMs · Vector Databases

Machine Learning Intern

09/2023 – 07/2024

UniQreate · Remote

- Built and deployed a production RAG system for document intelligence using LLMs, vector databases, and serverless cloud infrastructure on Azure
- Designed data extraction pipelines with rapid versioning, architecture design, and production testing cycles

Python · PyTorch · Azure (Blob + Serverless) · RAG · LLMs · Vector DBs

Undergraduate Research Assistant

10/2022 – 06/2025

UC San Diego · Dr. Mary Boyle's Lab · San Diego, CA

- Applied statistical modeling (regression, causal inference) to analyze UK BioBank and ABCD Study datasets for relationships between biomarkers and neuroimaging outcomes
- Developed reproducible analysis pipelines in Python using NumPy, Pandas, and Statsmodels for epidemiology research

Python · NumPy · Pandas · Statsmodels · Statistical Modeling · Epidemiology

RESEARCH & PROJECTS

Portfolio Allocation as Layered System Optimization

End-to-end quantitative research pipeline for portfolio construction. Designed a walk-forward backtesting engine with convex quadratic programming (Markowitz with ℓ_1 turnover penalty), regime detection via statistical jump models, and a policy-gradient reinforcement learning router for dynamic strategy selection. Modeled multi-agent competition for liquidity as a Nash equilibrium problem using iterative best-response with crowding penalties. Validated across top-100, top-250, and top-500 equity universes with provenance-ledger tracking to eliminate look-ahead bias.

Tech: Python, cvxpy, NumPy/SciPy, PyTorch, PufferLib, Nautilus Trader, D3.js

Research presentation ↗ · Interactive D3 demo ↗ · github.com/ashesh8500/fractal ↗

zkPHIRE: Zero-Knowledge Proof Accelerator on FPGA

Implemented a paper-parity hardware accelerator for the zkPHIRE SumCheck protocol on Xilinx PYNQ-Z2 (Zynq-7020). Designed Montgomery multiplier units to reduce DSP utilization from 200+ to 8-12 per multiplier. Built a fused update+extension pipeline with banked scratchpads for memory-efficient MLE evaluation. Optimized resource allocation to fit within 220 DSP and 280 BRAM constraints using Vitis HLS and

SystemVerilog.
Tech: Vitis HLS, SystemVerilog, C++, Python, Vivado, FPGA (Xilinx Zynq-7020)
Independent research project — paper reference: zkPHIRE (HPCA 2026)

RL-Based Autonomous Driving

Deep reinforcement learning for autonomous driving using the CARLA simulator. Implemented and compared policy gradient methods, DQN variants, and sim-to-real transfer strategies with real-world sensor data integration.
Tech: Python, PyTorch, CARLA, OpenAI Gym
github.com/ashesh8500/fp185 ↗

MediaSync — High-Performance Media Pipeline

Local-first media library pipeline: content-aware deduplication with SHA-256, hardware-accelerated transcoding to MP4, and batched cloud upload. Built in Rust with a terminal UI for large-scale photo/video library management.
Tech: Rust, FFmpeg, rclone, terminal UI
github.com/ashesh8500/mediasync ↗

Personal Website with On-Device AI Companion

Portfolio site with a browser-native AI assistant running a 1-bit quantized Bonsai-1.7B ONNX model on WebGPU. Built a Cloudflare Worker proxy for DeepSeek API with rate limiting and SSE streaming. Inference toggle between local WebGPU and cloud models.
Tech: JavaScript, ONNX Runtime Web, WebGPU, Cloudflare Workers, Transformers.js
asheshkaji.com ↗

TECHNICAL SKILLS

Languages:

Python | Rust | C/C++ | SystemVerilog | JavaScript | SQL | Bash

ML & Quantitative Research:

PyTorch | Scikit-Learn | NumPy/SciPy | Pandas | cvxpy | Statsmodels | PufferLib (RL) | Nautilus Trader | Reinforcement Learning | Convex Optimization | Walk-Forward Backtesting | D3.js

FPGA & Hardware:

Vitis HLS | Vivado | Xilinx Zynq | FPGA Design | ONNX Runtime | WebGPU

Infrastructure & MLOps:

Docker | Git | AWS | Azure | Cloudflare Workers | LangChain | AI-SDK | MCP/ACP | RAG | Vector Databases | LLMs | GitHub Actions

LANGUAGES

English — Native/Bilingual | Gujarati — Native/Bilingual | Hindi — Native/Bilingual